Research Methodology in Public Administration

Proposal Writing and Report Writing

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Presentation Outline

What We studied in Sem I

Syllabus Covered in Sem I

Philosophy of Social Science & Qualitative Methods

- Philosophy of social science
- Rationalism, Empericism & Sci. Method
- Theory of Causality
- Karl Popper's Principle
- Khun's Paradigm Shift
- Social Science and Natural Science
- Explanation of Human Action
- Phenomenology & Scoial Science
- Critical Theory
- Interpretation of Human Action

Data Analysis & Statistical Tools

- Descriptive statistics
- Probability & Probability Theory
- Estimation & Test of Hypothesis
- Analysis of Variance
- Simple Regression analysis
- Multiple Regression analysis
- Multivariate Analysis
- Analysis of Contingency Tables
- Non Parametric Methods
- Statistical software/tools

Building on Your Previous Knowledge (Sem I)

Philosophy of Social Science & Qualitative Methods

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From Knowledge to Research

Semester we have achieved the foundation of Research.

Semester II: Where Are We Going?

Proposal Writing $\longrightarrow$ Research Process $\longrightarrow$ Report Writing

What You Already Know

Research Philosophy + Qualitative Methods + Data Analysis

$\Downarrow$

What You Will Learn This Semester

How to transform an **idea or research problem** into a

Research Proposal and Research Report

R. M. in Public Administration III

Proposal Writing and Report Writing

Semester II

Course Information

Course Details

Course No.:	503
Credit Hours:	3
Title:	Research Methodology in Public Administration III
Focus:	Proposal Writing and Report Writing
Course Type:	Core & Compulsory

Course Objective

The objective of this course is to develop knowledge and skills on:

- Designing and writing an academic research proposal.
- Selecting appropriate research methods and designs.
- Collecting and analyzing research data.
- Conducting academic research systematically.
- Writing and presenting a scientific research report.

Course Contents: Foundations of Social Research

1. Overview of Scientific Method and Social Inquiry
2. Varieties of Social Research and Research Process
3. Dimensions and Major Types of Social Research
4. Methodology and Research Philosophies
    - Philosophy and Research Approaches
    - Feminist Research
    - Postmodern Research
5. The Literature Review

Course Contents: Research Design

6. Research Design

- Quantitative Research Design
- Qualitative Research Design
- Mixed Methods Research Design
- Causal Hypothesis and Testing
- Potential Errors in Causal Hypothesis:
  - Tautology
  - Teleology
  - Ecological Fallacy
  - Reductionism
  - Spuriousness
- Qualitative Research Design

Course Contents: Measurement and Sampling

7. Quantitative and Qualitative Measurement
    - Conceptualization and Operationalization
    - Validity and Reliability
    - Levels of Measurement
    - Scores, Indexes and Scales
8. Quantitative and Qualitative Sampling
    - Probability Sampling
    - Non-probability Sampling

Course Contents: Data Collection

9. Data Collection Instruments

- Questionnaire
- Interview
- Observation and its Types
- Participatory Rapid Appraisal
- Ethnography
- Focus Group Discussion
- Case Studies
- Survey Research

10. Survey Research
    - Non-reactive Research
    - Field Research

Course Contents: Data Analysis

11. Analysis of Data

Quantitative Data

- Univariate Analysis
- Bivariate Analysis
- Multivariate Analysis

Qualitative Data

- Coding
- Categorization
- Thematic Analysis
- Interpretation

Major Components of the Course

1. Ethics in Social Research
2. Writing of Research Proposal
3. Conducting Academic Research
4. Analysis and Interpretation of Data
5. Scientific Report Writing

Research Idea → Proposal → Research → Report

Model Formulation

Consider the general ARIMA model

$$\phi(B)(1 - B)^d Y_t = \theta(B)\varepsilon_t,$$

where

- B is the backshift operator,
- d is the order of differencing,
- $\phi(B)$ represents the autoregressive component,
- $\theta(B)$ represents the moving-average component.

Results

Model Selection

Model	AIC	BIC	RMSE
ARIMA(1,1,1)	425.3	438.2	12.41
ARIMA(2,1,1)	421.7	435.8	11.83
ARIMA(1,1,2)	423.5	437.1	12.06

Key Result

ARIMA$(2, 1, 1)$ provides the best overall model performance.

Forecasting Results

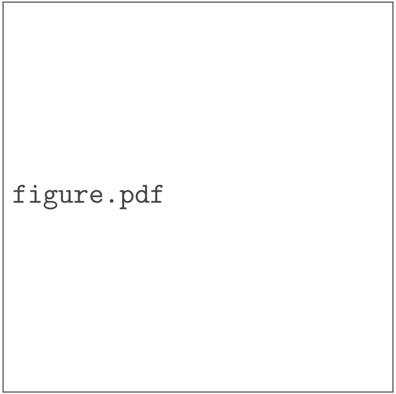

Figure: Observed and forecast values

Graphical Results

figure1.pdf

figure2.pdf

Discussion

Discussion

Major Findings

- Strong temporal dependence
- Appropriate differencing
- Good forecasting accuracy

Implications

- Useful for planning
- Supports decision making
- Provides future estimates

Conclusion

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The proposed statistical model provides a useful framework for describing the observed pattern and forecasting future values.

- Model identification was performed systematically.
- Diagnostic tests supported model adequacy.
- Forecasts can be used for future planning.

Thank You

Questions and Discussion

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